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GCSE BIOLOGY

TOPIC 3: INFECTION AND RESPONSE

Detailed AQA Specification Notes

Clear explanations | HT content | Exam tips | Revision checklist

HOW TO USE THIS BOOKLET

Learn each section, then cover the notes and explain the biology aloud. Use the disease tables to practise pathogen, transmission, symptoms and control. The content follows the uploaded AQA Topic 3 specification and expands each stated point to an exam ready GCSE level.

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Topic map

Specification focus: everything in the uploaded Infection and Response specification extract

Section	Focus	What you must know
1	Pathogens and transmission	Pathogen groups, how they cause disease and how infection spreads
2	Viral diseases	Measles, HIV and tobacco mosaic virus
3	Bacterial diseases	Salmonella food poisoning and gonorrhoea
4	Fungal and protist diseases	Rose black spot and malaria
5	Human defence systems	Skin, nose, trachea, bronchi and stomach
6	The immune system	Phagocytosis, antibodies and antitoxins
7	Vaccination	Individual protection and reducing spread through population immunisation
8	Antibiotics and medicines	Bacterial treatment, resistance, painkillers and viral disease
9	Drug development	Origins, preclinical testing, clinical trials, placebo and peer review
10	Monoclonal antibodies	Production from hybridomas and single clones
11	Uses of monoclonal antibodies	Diagnosis, research, measurement and targeted treatment
12	Plant disease detection	HT symptoms and methods used to identify plant disease
13	Plant disease and deficiencies	TMV, black spot, aphids, nitrate and magnesium deficiency
14	Plant defence responses	Physical, chemical and mechanical plant defences
15	Exam skills and checklist	Comparisons, evaluation points and full topic revision checklist

SPECIFICATION FLAGS

HT ONLY is used only where the uploaded specification explicitly labels content as Higher Tier. No required practical activity is specified in this Topic 3 extract.

1 Pathogens and transmission

Specification focus: viruses, bacteria, protists and fungi; spread and prevention

A pathogen is a microorganism that causes infectious disease. The four pathogen groups named in the specification are viruses, bacteria, protists and fungi. Pathogens can infect animals or plants.

Pathogen group	How it causes disease / key feature	Example from this topic
Virus	Lives and reproduces inside cells, causing cell damage.	Measles, HIV, tobacco mosaic virus
Bacterium	May reproduce rapidly in the body and may release toxins that damage tissues.	Salmonella, gonorrhoea
Protist	A disease causing microorganism; some use a vector as part of their life cycle.	Malaria
Fungus	Can infect plant tissues and reduce growth.	Rose black spot

How pathogens can spread

Route	What it means	Topic example
Direct contact	Infection passes directly between organisms.	Sexual contact can spread HIV and gonorrhoea.
Air / droplets	Pathogens travel in droplets released from coughs or sneezes.	Measles.
Food	Pathogens are swallowed in contaminated food or food prepared unhygienically.	Salmonella.
Water / wind	Plant pathogens can be carried through the environment.	Rose black spot.
Vector	Another organism carries the pathogen between hosts.	Mosquitoes transmit the malaria protist.

EXAM TIP

Do not write that all pathogens spread in the same way. Match the route of transmission and the control method to the specific disease in the question.

2 Viral diseases

Specification focus: measles, HIV and tobacco mosaic virus

Disease	Transmission	Key symptoms / effects	Control or treatment named in the specification
Measles	Inhalation of droplets from coughs and sneezes.	Fever and a red skin rash. It can be serious and fatal if complications arise.	Most young children are vaccinated against measles.
HIV	Sexual contact or exchange of body fluids such as blood, including shared needles.	Initially a flu-like illness. The virus attacks immune cells. Late stage infection (AIDS) leaves the immune system unable to deal with other infections or cancers.	Antiretroviral drugs can control the infection.
Tobacco mosaic virus (TMV)	A widespread plant viral pathogen.	A mosaic pattern of leaf discolouration. Reduced photosynthesis causes reduced plant growth.	The uploaded extract focuses on recognition and effects rather than a named treatment.

Why viruses can be difficult to treat

Viruses reproduce inside body cells. This makes drug treatment difficult because a drug that damages the virus may also damage the host tissues. The specification therefore states that it is difficult to develop drugs that kill viruses without also damaging body tissues.

CAUSE AND EFFECT CHAIN FOR TMV

TMV infects the plant -> leaves become discoloured -> less photosynthesis occurs -> less glucose is produced for growth -> the plant grows less well.

COMMON MISTAKE

Antibiotics do not kill viral pathogens. Do not suggest antibiotics as a treatment for measles, HIV or TMV.

3 Bacterial diseases

Specification focus: salmonella food poisoning and gonorrhoea

Salmonella food poisoning

- Spread occurs when Salmonella bacteria are ingested in contaminated food or food prepared in unhygienic conditions.
- Symptoms include fever, abdominal cramps, vomiting and diarrhoea.
- The symptoms are caused by the bacteria and the toxins they secrete.
- In the UK, poultry are vaccinated against salmonella to help control its spread.

Gonorrhoea

- Gonorrhoea is a sexually transmitted disease caused by a bacterium.
- Symptoms include a thick yellow or green discharge from the vagina or penis and pain when urinating.
- It spreads by sexual contact.
- The spread can be controlled by antibiotic treatment or by using a barrier method of contraception such as a condom.
- Gonorrhoea was once easily treated using penicillin, but many resistant strains have appeared.

EXAM COMPARISON

Both salmonella and gonorrhoea are bacterial diseases, but their transmission is different. Salmonella is associated with contaminated food; gonorrhoea is spread by sexual contact.

4 Fungal and protist diseases

Specification focus: rose black spot and malaria

Rose black spot

- Rose black spot is a fungal disease of plants.
- Purple or black spots develop on the leaves. The leaves may turn yellow and drop early.
- Loss of healthy leaf area reduces photosynthesis, so plant growth is reduced.
- The disease is spread through the environment by water or wind.
- It can be treated using fungicides and/or by removing and destroying affected leaves.

Malaria

- Malaria is caused by a protist.
- The malarial protist has a life cycle that includes a mosquito. The mosquito therefore acts as a vector.
- Malaria causes recurrent episodes of fever and can be fatal.
- The spread can be reduced by preventing mosquitoes from breeding and by using mosquito nets to avoid bites.

VECTOR DEFINITION

A vector is an organism that carries a pathogen from one host to another. In malaria, the mosquito carries the protist between people.

EXAM TIP

For plant disease questions, link damage to photosynthesis when the specification gives you a leaf symptom. Less healthy leaf area means less photosynthesis and therefore reduced growth.

5 Human non-specific defence systems

Specification focus: skin, nose, trachea, bronchi and stomach

Non-specific defences act against many different pathogens. They reduce the chance of pathogens entering the body or surviving long enough to establish an infection.

Defence	How it protects the body
Skin	Forms a physical barrier that makes it difficult for pathogens to enter the body.
Nose	Helps trap particles and microorganisms before they travel deeper into the respiratory system.
Trachea and bronchi	Mucus can trap pathogens and particles; movement of mucus helps remove them from the airways.
Stomach	Acidic conditions help destroy many microorganisms that are swallowed.

Why these are called non-specific

They do not target one particular antigen or one particular pathogen. Their role is to provide a general first line of defence.

EXAM COMMAND WORDS

If asked to describe, state what the defence does. If asked to explain, link the structure or condition to how it prevents pathogen entry or survival.

6 The immune system

Specification focus: phagocytosis, antibody production and antitoxin production

If a pathogen enters the body, the immune system attempts to destroy it. White blood cells are central to this response.

White blood cell response	What happens
Phagocytosis	A white blood cell engulfs a pathogen and digests it.
Antibody production	White blood cells produce antibodies. The antibodies are specific to particular antigens and help target the pathogen.
Antitoxin production	White blood cells produce antitoxins that neutralise toxins released by bacteria.

A useful exam sequence

1. A pathogen enters the body.
2. White blood cells recognise that it is foreign.
3. White blood cells can engulf pathogens, make specific antibodies and produce antitoxins.
4. The pathogen or its harmful products are destroyed or neutralised.

PRECISION MATTERS

Antibodies act against specific antigens. Antitoxins neutralise toxins. Do not use the two words as if they mean the same thing.

7 Vaccination

Specification focus: preventing illness in an individual and reducing pathogen spread

Vaccination prepares the immune system so that it can respond much more quickly if the same pathogen enters the body later.

5. Small quantities of dead or inactive forms of a pathogen are introduced into the body.
6. White blood cells are stimulated to produce antibodies.
7. If the same pathogen enters the body again, the white blood cells respond quickly.
8. The correct antibodies are produced rapidly, preventing infection from developing into illness.

Why vaccinating many people reduces spread

When a large proportion of the population is immunised, there are fewer susceptible people for a pathogen to infect and pass between. This reduces the overall spread of the pathogen.

Individual benefit	Population benefit
The vaccinated person develops a faster specific immune response to that pathogen.	Widespread immunisation reduces opportunities for the pathogen to move between hosts.

EVALUATION POINT

The specification expects you to be able to evaluate the global use of vaccination. In an evaluation, weigh the disease prevention benefit against practical issues involved in immunising large populations, using any information provided in the question.

NOT REQUIRED

You do not need to recall detailed vaccination schedules or specific vaccine side effects for this specification point.

8 Antibiotics and other medicines

Specification focus: bacterial disease, resistance, viral disease and symptom relief

Medicine / issue	Specification knowledge
Antibiotics	Medicines such as penicillin that help cure bacterial disease by killing infective bacteria inside the body.
Specific antibiotics	Specific bacteria need to be treated using suitable specific antibiotics.
Antibiotic resistance	The emergence of resistant bacterial strains is a major concern and reduces the effectiveness of antibiotics.
Viruses	Antibiotics cannot kill viral pathogens.
Painkillers and other medicines	Can treat symptoms, but do not kill the pathogen causing the disease.
Antiviral drug difficulty	It is difficult to kill viruses without also damaging body tissues because viruses live and reproduce inside cells.

Why antibiotics were important

The use of antibiotics greatly reduced deaths from infectious bacterial diseases. The specification also emphasises that antibiotic resistant strains are now a serious concern.

COMMON EXAM MISTAKE

“The patient feels better” does not prove that a medicine killed a pathogen. A painkiller may reduce pain or fever while the pathogen remains in the body.

9 Discovery and development of medicines

Specification focus: origins, testing, clinical trials and peer review

Where medicines can come from

Example	Origin named in the specification
Digitalis	Foxgloves
Aspirin	Willow
Penicillin	Discovered by Alexander Fleming from Penicillium mould

Most new drugs are synthesised by chemists in the pharmaceutical industry, although the starting point may still be a chemical extracted from a plant.

What new drugs must be tested for

Term	Meaning for drug testing
Toxicity	Is the drug harmful or poisonous?
Efficacy	Does the drug produce the intended medical effect?
Dose	What amount gives the best balance of effect and safety?

Preclinical testing

Preclinical testing takes place in laboratories using cells, tissues and live animals before clinical trials are carried out in people.

THREE WORDS TO REMEMBER
TOXICITY | EFFICACY | DOSE

10 Clinical trials and peer review

Specification focus: healthy volunteers, patients, placebo, double blind trials and scrutiny

9. Clinical trials use healthy volunteers and patients.
10. Very low doses are given at the start of a clinical trial.
11. If the drug is found to be safe, further trials are carried out to find the optimum dose.
12. In double blind trials, some patients are given a placebo.
13. Results are published only after scrutiny by peer review.

Why use a placebo?

A placebo gives a comparison group that has not received the active drug. This helps researchers judge whether changes are due to the drug rather than simply to expectations or other factors.

Why use a double blind design?

A double blind trial reduces bias because neither the participant nor the person assessing the treatment knows which participants received the real treatment during the trial.

Why peer review matters

Peer review means other scientists scrutinise the work before it is accepted. The specification expects you to understand that trial results are published only after this scrutiny.

EXAM TIP

When comparing two treatments, look for sample size, whether a control or placebo was used, whether the design was blinded, and whether the evidence actually shows a meaningful difference.

11 Producing monoclonal antibodies

Specification focus: single clones, mouse lymphocytes and hybridoma cells

Monoclonal antibodies are produced from a single clone of cells. They are specific to one binding site on one protein antigen, allowing them to target a specific chemical or specific cells.

Production sequence

14. Mouse lymphocytes are stimulated to make the required antibody.
15. The lymphocytes are combined with a particular type of tumour cell.
16. The combined cell is called a hybridoma.
17. A hybridoma can both divide and make the antibody.
18. Single hybridoma cells are cloned to produce many identical cells.
19. All cells in the clone produce the same antibody.
20. Large quantities of the antibody can then be collected and purified.

WHY USE A HYBRIDOMA?

A lymphocyte makes the desired antibody. A tumour cell can divide repeatedly. Combining them produces a hybridoma that can both divide and make the antibody.

KEY WORD

MONOCLONAL means the antibodies come from one clone and are therefore identical in their antigen specificity.

12 Uses of monoclonal antibodies

Specification focus: diagnosis, measurement, research and targeted treatment

Use	How monoclonal antibodies are used
Diagnosis	Used in tests such as pregnancy tests.
Measuring chemicals	Used in laboratories to measure levels of hormones and other chemicals in blood, or to detect pathogens.
Research	Bound to a fluorescent dye so that specific molecules in cells or tissues can be located or identified.
Treating disease	For cancer, an antibody can be attached to a radioactive substance, toxic drug or chemical that stops cells growing and dividing. The antibody delivers it to cancer cells while reducing harm to other cells.

What you are expected to do in an exam

You are not expected to memorise specific monoclonal antibody tests or treatments. Instead, if you are given information, you should be able to explain how the antibody works by binding specifically to its target.

Evaluation

- Monoclonal antibodies can target specific cells or chemicals, which gives them powerful diagnostic and treatment uses.
- The specification states that monoclonal antibodies created more side effects than expected.
- They are not yet as widely used as people originally hoped.
- You should be able to evaluate advantages, disadvantages and ethical issues using information supplied in a question.

EXAM CHAIN

specific antibody -> binds to target antigen -> attached marker or treatment is delivered to that target -> other cells are less affected

13 Plant disease detection

Specification focus: HT ONLY: symptoms and methods of identification

HIGHER TIER ONLY
The detection and identification methods on this page are explicitly labelled HT only in the uploaded specification.

Signs that can indicate plant disease

Possible symptom	What you may observe
Stunted growth	The plant grows less than expected.
Spots on leaves	Unusual areas or lesions appear on leaf surfaces.
Decay / rot	Areas of tissue break down.
Growths	Abnormal lumps or growths develop.
Malformed stems or leaves	Plant structures develop an abnormal shape.
Discolouration	Leaves or other tissues change colour.
Presence of pests	Insects or other pests are visible on the plant.

How the cause can be identified

- Compare symptoms with a gardening manual or website.
- Take infected plant material to a laboratory to identify the pathogen.
- Use testing kits that contain monoclonal antibodies.

APPLICATION SKILL
A symptom tells you that a plant may be diseased, but it does not always identify the exact cause. Identification methods provide more specific evidence.

14 Plant diseases and mineral ion deficiencies

Specification focus: TMV, black spot, aphids, nitrate deficiency and magnesium deficiency

Plants can be infected by viral, bacterial and fungal pathogens and can also be damaged by insects. The specification restricts named plant disease knowledge to TMV, black spot and aphids as insects.

Problem	Type	Effect / key knowledge
Tobacco mosaic virus	Viral disease	Mosaic leaf discolouration reduces photosynthesis and growth.
Black spot	Fungal disease	Dark leaf spots, yellowing and early leaf loss reduce photosynthesis and growth.
Aphids	Insect pest	Aphids are the named insect pest in the specification.
Nitrate deficiency	Mineral ion deficiency	Causes stunted growth because nitrate ions are needed for protein synthesis and therefore growth.
Magnesium deficiency	Mineral ion deficiency	Causes chlorosis because magnesium ions are needed to make chlorophyll.

Two deficiency chains to learn

NITRATE DEFICIENCY
less nitrate -> less protein synthesis -> poorer growth -> stunted plant

MAGNESIUM DEFICIENCY
less magnesium -> less chlorophyll -> chlorosis (yellow leaves)

Understanding ion deficiencies allows horticulturists to provide plants with more suitable growing conditions.

15 Plant defence responses

Specification focus: physical, chemical and mechanical defences

Defence type	Examples named in the specification	How it helps
Physical	Cellulose cell walls; tough waxy cuticle; layers of dead cells around stems such as bark.	Creates barriers that resist invasion by microorganisms.
Chemical	Antibacterial chemicals; poisons that deter herbivores.	Kills or inhibits microorganisms, or discourages animals from feeding.
Mechanical adaptations	Thorns and hairs; leaves that droop or curl when touched; mimicry.	Makes feeding more difficult or tricks/deters animals.

Bark as a defence

Layers of dead cells around stems form a protective barrier. These layers can also fall off, helping remove microorganisms from the plant surface.

EXAM TIP

If asked to classify a defence, first decide whether it is a barrier (physical), a substance (chemical), or a structure/response that deters herbivores (mechanical adaptation).

16 High value comparison grids

Specification focus: fast recall for common exam questions

Pathogen comparison

Pathogen type	Named example	Key effect
Virus	Measles / HIV / TMV	Reproduces inside cells and damages them.
Bacterium	Salmonella / gonorrhoea	Can reproduce rapidly and may release toxins.
Fungus	Rose black spot	Damages leaves and reduces photosynthesis.
Protist	Malaria	Uses a mosquito vector as part of its life cycle.

Treatment and prevention comparison

Approach	What it does	What it does not do
Vaccination	Stimulates a specific immune response before later exposure.	Does not directly kill an existing pathogen.
Antibiotic	Kills infective bacteria when a suitable antibiotic is used.	Does not kill viruses.
Painkiller	Reduces symptoms such as pain.	Does not kill the pathogen.
Barrier contraception	Reduces direct sexual transmission of diseases such as gonorrhoea.	Is not an antibiotic treatment.
Mosquito nets / breeding control	Reduces contact with the malaria vector.	Does not act as an antibiotic.

MEMORY CHECK

Can you give the pathogen type, transmission route, main effect and one control method for each named disease without looking?

17 Exam practice

Specification focus: short answer, explain and evaluate questions

Quick recall

21. Define a pathogen. [1]
22. Name the four types of pathogen in this topic. [4]
23. Explain why TMV can reduce plant growth. [3]
24. Describe three ways white blood cells defend the body. [3]
25. Explain how vaccination can prevent illness when the same pathogen enters the body later. [4]
26. Explain why antibiotics do not cure viral diseases. [2]
27. Name the three factors new drugs are tested for. [3]
28. Describe how a hybridoma is produced and used to make monoclonal antibodies. [5]
29. Give two symptoms that could indicate plant disease. [2]
30. Explain why magnesium deficiency causes chlorosis. [2]

Longer exam prompts

- Compare the transmission and control of measles, gonorrhoea and malaria. [6]
- Explain how the body prevents pathogens entering and how the immune system responds if they do enter. [6]
- Evaluate the use of monoclonal antibodies for treating disease using information provided in a question. [6]
- Describe the stages used to test a new medicine before it becomes widely available. [6]

SIX MARK STRUCTURE

Use linked biological statements rather than disconnected facts. For evaluate questions, include both benefits and limitations, then use the information in the question to reach a reasoned judgement.

18 Final revision checklist

Specification focus: what you should be able to explain without notes

☐ Define pathogen and name viruses, bacteria, protists and fungi as pathogen groups.

☐ Explain direct contact, air/droplets, food, environmental spread and vector transmission using topic examples.

☐ Recall the pathogen, symptoms/effects, transmission and control of measles, HIV, TMV, salmonella, gonorrhoea, rose black spot and malaria.

☐ Describe the non-specific defences of the skin, nose, trachea/bronchi and stomach.

☐ Explain phagocytosis, antibody production and antitoxin production.

☐ Explain how vaccination protects an individual and why immunising a large proportion of a population reduces spread.

☐ Explain the use and limitations of antibiotics, painkillers and medicines for viral disease.

☐ Describe the origin and testing of new drugs, including toxicity, efficacy, dose, preclinical testing, clinical trials, placebo, double blind trials and peer review.

☐ Describe how monoclonal antibodies are produced from mouse lymphocytes and tumour cells using hybridomas.

☐ Explain uses of monoclonal antibodies in diagnosis, measurement, research and targeted treatment.

☐ HT: recognise signs of plant disease and describe the three identification methods in the specification.

☐ Recall TMV, black spot and aphids as the restricted named plant examples.

☐ Explain nitrate deficiency and magnesium deficiency.

☐ Describe physical, chemical and mechanical plant defence responses.

FINAL EXAM HABIT

For every named disease, practise four headings: PATHOGEN | TRANSMISSION | EFFECT/SYMPTOMS | CONTROL. For mechanisms such as vaccination or monoclonal antibodies, practise writing the steps in a logical order.

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